

Distributing Replicas

1. What is Distributing Replicas?

Distributing replicas means storing copies of data across different physical machines, racks, availability zones, or data centers.

None

Primary Copy

└─ Replica on same rack

└─ Replica on different rack

└─ Replica in another region

Goal:

- High availability
- Fault tolerance
- Disaster recovery
- Low latency access

2. Typical Replica Placement Strategy

A common production setup:

A. Same Rack Replica

Fast network, low latency.

Used for quick failover.

B. Different Rack Replica

Protects against:

- Rack power failure
- Top-of-rack switch failure
- Hardware group outage

C. Different Data Center / Region Replica

Protects against:

- Zone outage
- Flood/fire/power loss
- Major disaster

3. Why Multiple Levels Matter

If all replicas are in one rack:

None

`Rack fails = all replicas fail`

So distribute across failure domains.

4. What is a Failure Domain?

A failure domain is any unit that can fail together.

Examples:

- Disk
- Server
- Rack
- Availability Zone
- Region

Good HLD design spreads replicas across domains.

5. Example Architecture

None

`Mumbai Zone A → Primary`

`Mumbai Zone B → Replica`

`Singapore Region → Replica`

Now even if Zone A crashes, system survives.

6. Sharding + Replication Together

Large systems use both:

Sharding

Split data into chunks.

Replication

Copy each shard.

Example:

None

Shard 1 → 3 replicas

Shard 2 → 3 replicas

Shard 3 → 3 replicas

Used by:

- MongoDB
- Apache Cassandra
- Elasticsearch

7. Benefits of Sharding Mentioned

A. Scale Storage

If one machine cannot hold full database.

None

10 TB data → split across 10 nodes

B. Scale Memory

In-memory DBs like Redis may shard because RAM is expensive.

C. Scale Processing

Parallel queries across shards.

D. Data Locality

Store user data near relevant services/users.

Example:

None

Indian users on India shard

EU users on EU shard

Lower latency.

8. Trade-offs of Sharding

Main cost:

None

More complexity

Need to manage:

- Shard mapping
- Rebalancing
- Cross-shard joins
- Hot partitions
- Failover per shard

9. Tombstones / Soft Deletes

Instead of immediate delete, some distributed systems mark records deleted.

None

`DELETE user123`

→ `Write tombstone marker`

Benefits:

- Ongoing reads not broken
- Replicas sync safely
- Eventual consistency easier

Used in systems like:

- Apache Cassandra
- Apache Hadoop ecosystems

10. Interview HLD Talking Points

When asked about replica distribution discuss:

- Replica count (RF=3 common)
- Same AZ vs cross AZ
- Cross-region DR replica
- Sync vs async replication
- Failover time
- Data sovereignty laws
- Cost vs reliability tradeoff

11. Example Decision

E-commerce Orders DB

None

Primary: Mumbai

Replica 1: Mumbai AZ-2

Replica 2: Singapore

Why:

- Local low latency writes
- Zone failover
- Regional DR

12. Short Revision Notes

None

Distribute replicas across:

server

rack

zone

region

Use sharding for:

storage

RAM

parallelism

locality

13. One-Line Summary

Distributing replicas across multiple failure domains and combining with sharding enables scalable, fault-tolerant, high-availability databases.